Hyperspherical Bessel functions in CAMB: Olver mapping, recurrence fallback, shifted-ν, and Numerov source integration

Implementation note for CAMB non-flat Bessel calculations

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Abstract

This note explains how CAMB evaluates the regular hyperspherical Bessel functions (also called ultraspherical Bessel functions) used in non-flat source integration. The mathematical object is the curved-space radial function $\phi_l^\nu(K,\chi)$, whose flat limit is the spherical Bessel function $j_l(\nu\chi)$. The main fast point approximation is a leading-order Olver, or uniform Liouville-Green, mapping: the curved radial equation is transformed into a flat spherical-Bessel equation at an effective radius $z(\chi)$, with a transport amplitude $A(\chi)$. This makes the calculation exact in the flat limit and smooth through the turning point, but it does not evaluate the higher Olver correction terms. The note also compares this with earlier Langer/Airy WKB and CLASS flat-rescaling approaches, explains the stable recursive fallback `phi_recurs` and the rare open small-ν fallback, the closed-space Gegenbauer Miller start, how CAMB obtains nearby Bessel values by Numerov integration during source integration, and how the current shifted-ν shortcut is the constant-map, fastest small-χ limit of the Olver expansion.

Contents

1 What is the function and why does CAMB need it?

In a curved Friedmann-Robertson-Walker spatial section, scalar harmonics separate into angular spherical harmonics and a radial function,

$$Q_{\nu lm}(\chi,\theta,\varphi) = \phi_l^\nu(K,\chi)Y_{lm}(\theta,\varphi). \tag{1}$$

The radial function ϕ_l^ν is the hyperspherical Bessel function. It is the non-flat generalization of the ordinary spherical Bessel function used in line-of-sight integration. These definitions and normalizations follow the standard curved-space CMB convention [3, 4, 5, 6]. CAMB has to evaluate it many times for many source times, many wavenumbers and many multipoles, so direct special-function evaluation at every source point would be too slow.

The dimensional FRW spatial curvature will be denoted by $\mathcal{K}$. It has units of inverse length squared and is not restricted to ±1. For the dimensionless Olver equations used in the Bessel routines, distances are rescaled by

$$R_K = |\mathcal{K}|^{-1/2}, \qquad \chi = \frac{r}{R_K}, \qquad \mathcal{K} \neq 0, \tag{2}$$

and only the sign remains:

$$K = \mathrm{sgn}(\mathcal{K}) = +1, 0, -1, \tag{3}$$

for closed, flat and open spaces. Equivalently, $\mathcal{K} = K/R_K^2$ for $K = \pm 1$, with the flat case obtained as the limit $\mathcal{K} \to 0$. The radial metric function in the dimensionless coordinate χ is

$$S_K(\chi) = \begin{cases} \sin\chi, & K = +1, \\ \chi, & K = 0, \\ \sinh\chi, & K = -1, \end{cases} \qquad C_K(\chi) = \frac{\mathrm{d}S_K}{\mathrm{d}\chi} = \begin{cases} \cos\chi, & K = +1, \\ 1, & K = 0, \\ \cosh\chi, & K = -1. \end{cases} \tag{4}$$

We also use

$$\cot_K \chi = \frac{C_K(\chi)}{S_K(\chi)}. \tag{5}$$

Thus $\cot_K \chi$ means $\cot\chi$, $1/\chi$, or $\coth\chi$ in the closed, flat or open cases.

The symbols q, ν, and k are not interchangeable in all curvature conventions. The transfer-grid variable q is dimensional. In a non-flat Bessel call the dimensionless radial eigenvalue is

$$\nu = qR_K, \tag{6}$$

which is also the argument name used by the low-level Bessel routines. For scalar spectra the primordial power is evaluated at the physical wavenumber

$$k^2 = q^2 - \mathcal{K}, \tag{7}$$

so, in dimensionless form, $(kR_K)^2 = \nu^2 - K$. For tensors the corresponding relation is $k^2 = q^2 - 3\mathcal{K}$. In the flat limit $q = k$; away from flatness, ν labels the radial hyperspherical-Bessel eigenvalue while k is the physical wavenumber used by the primordial spectrum.

CAMB distinguishes between the unreduced radial function and a reduced function,

$$u_l^\nu(K,\chi) = S_K(\chi)\phi_l^\nu(K,\chi). \tag{8}$$

The reduced function is the natural one for asymptotics and Numerov integration because it obeys a second-order equation without a first-derivative term. The functions in the code therefore appear in two forms:

$$\texttt{phi\_olver}(l, K, \nu, \chi) \longrightarrow \phi_l^\nu(K,\chi), \tag{9}$$

$$\texttt{u\_olver}(l, K, \nu, \chi) \longrightarrow u_l^\nu(K,\chi). \tag{10}$$

2 Differential equation and normalization

The unreduced equation is

$$\phi'' + 2\cot_K \chi\, \phi' + \left[\nu^2 - K - \frac{l(l+1)}{S_K^2(\chi)}\right]\phi = 0. \tag{11}$$

Multiplying by S_K gives the reduced equation

$$u'' = \left[\frac{l(l+1)}{S_K^2(\chi)} - \nu^2\right] u. \tag{12}$$

Equivalently,

$$u'' + \left[\nu^2 - \frac{l(l+1)}{S_K^2(\chi)}\right] u = 0. \tag{13}$$

This is the equation used for Numerov stepping in the source integration code. It is also the Schrödinger-form equation used by the earlier Langer/WKB treatments of hyperspherical Bessel functions [4, 5]; CAMB's new approximation changes the comparison solution, not the underlying radial equation.

The regular solution is fixed by its behaviour at the origin. Define

$$b_j(K,\nu) = \sqrt{\nu^2 - Kj^2}. \tag{14}$$

Then

$$\phi_l^\nu(K,\chi) \sim \frac{\prod_{j=1}^{l} b_j(K,\nu)}{(2l+1)!!}\,\chi^l, \qquad \chi \to 0. \tag{15}$$

For $K=0$, this becomes

$$j_l(\nu\chi) \sim \frac{(\nu\chi)^l}{(2l+1)!!}. \tag{16}$$

For $K=+1$, ν is a discrete integer mode and the code requires $\nu > l$. In the closed case the coordinate is also folded into the fundamental interval, as explained below; the same closed-spectrum and symmetry constraints are discussed in the WKB and stable-recurrence references [4, 5]. The closed full interval is $0 < \chi < \pi$, so any statement about being "below" or "above" the turning point should be read in the folded coordinate unless the second closed turning point is explicitly being discussed.

3 The turning point and the qualitative regimes

The picture in this section is a large-l one: the Olver and WKB asymptotics treat $\ell = \sqrt{l(l+1)}$ as the large parameter, so the turning-point language and the numerical gates built from it apply to moderate and high multipoles, not to the lowest few. For $l \le 2$, and in the other difficult corners noted below, the code does not rely on this picture at all and falls back to the exact recurrence `phi_recurs`. With that caveat, let

$$L = l(l+1), \qquad \ell = \sqrt{L}, \qquad \alpha = \frac{\nu}{\ell}. \tag{17}$$

Then equation (13) can be written as

$$u'' + \ell^2\left[\alpha^2 - \frac{1}{S_K^2(\chi)}\right]u = 0. \tag{18}$$

The sign of the bracket tells us whether the regular solution is exponentially small or oscillatory. The turning point is where the bracket vanishes:

$$S_K(\chi_t) = \frac{1}{\alpha} = \frac{\ell}{\nu}. \tag{19}$$

Thus, using the $\ell = \sqrt{l(l+1)}$ convention used in the code paths discussed here,

$$\chi_t = \begin{cases} \operatorname{asin}(\ell/\nu), & K=+1,\\ \ell/\nu, & K=0,\\ \operatorname{asinh}(\ell/\nu), & K=-1. \end{cases} \tag{20}$$

This is the turning point of the reduced equation as written, with the exact $l(l+1)$ centrifugal coefficient. It is also the $\sqrt{l(l+1)}$ balance used in the CLASS hyperspherical WKB/Airy code and in the WKB formulae of Refs. [4, 5]. CAMB and other transfer codes also use the nearby $l+1/2$ convention in separate flat-Bessel peak, onset, or Limber estimates. The two conventions differ relative to l only at $O(l^{-2})$, but they are not the same object: the Olver action map in this note follows $l(l+1)$ because the comparison solution $zj_l(\nu z)$ satisfies the exact reduced flat spherical-Bessel equation with that coefficient.

For $K = +1$ on the full closed interval, $S_K(\chi) = \sin\chi$ gives two turning points, χ_t and $\pi - \chi_t$. The solution is evanescent near both closed poles and oscillatory between the two turning points. CAMB folds the argument into $[0, \pi/2]$, carrying the corresponding parity sign separately, so the local discussion in most of this note refers to the first turning point only: below it means the pre-peak or evanescent side, and above it means the oscillatory side. A direct WKB formula would have a singular-looking amplitude at the turning point. The true solution is finite, so the numerical method must be uniform through this transition.

The same parameter α is also a useful numerical gate. In the local small-χ curvature expansion the constant expansion parameter is

$$h = K\frac{l(l+1)}{\nu^2} = \frac{K}{\alpha^2}. \tag{21}$$

Thus larger α means both weaker curvature corrections in the local map and a smaller turning-point radius, $S_K(\chi_t) = 1/\alpha$. The near-flat table shortcuts use calibrated endpoint gates based on ν/l and the small-χ range metric $l^2\chi^7_{\max}/\nu$. The exact source-range gate is given once in (71); the shifted-ν path then adds the constant-map residual test described below.

This is also why the approximation is normally used in its comfortable regime for the near-flat cosmologies of most practical interest. If $|\mathcal{K}|^{-1/2}$ is much larger than the current horizon, then every source point in the line-of-sight integral has $\chi \lesssim \chi_0 = \tau_0\sqrt{|\mathcal{K}|} \ll 1$. The Bessel factor contributing to multipole l is largest near its peak or turning point, where $S_K(\chi) \simeq \ell/\nu$. Thus the contributing modes obey

$$\alpha = \frac{\nu}{\ell} \simeq \frac{1}{S_K(\chi)} \gtrsim \frac{1}{\chi_0} \gg 1, \tag{22}$$

or equivalently $\nu \gg l$ except in very low-l, endpoint, or exponentially small tail regions. Those exceptional corners are exactly where the code uses the calibrated gates, Numerov re-seeding, or stable recurrence rather than blindly trusting a near-flat shortcut.

4 What "Olver" means in this calculation

Olver asymptotics are a systematic form of the Liouville-Green or WKB method for second-order differential equations with a large parameter and turning points [1]. The practical recipe is:

> Choose a comparison equation whose solution is already known, map the independent variable so that the action integral from the turning point agrees in the original and comparison equations, and include the corresponding amplitude transport factor.

For CAMB's default pointwise Olver path, the comparison equation is the *flat spherical-Bessel equation.* The reference solution is the reduced flat mode

$$v(z) = zj_l(\nu z),$$

so the implementation evaluates the curved function by mapping $\chi \mapsto z(\chi)$, applying the transport amplitude, and then calling the ordinary flat spherical-Bessel routine `BJL`. In this note, unless otherwise stated, "the Olver approximation" means this spherical-Bessel reference problem, not an Airy reference problem.

This should be separated from two related comparison choices in the older literature. The Langer/Airy uniform WKB approximation, used by Kosowsky and as a WKB reference in the later hyperspherical-Bessel papers [4, 5, 6], maps the curved radial equation directly to an Airy equation. The CLASS high-ν, near-flat rescaling approximation also avoids expensive hyperspherical evaluations by calling ordinary flat spherical Bessel functions at a mapped argument [6]. CAMB's default map is closest in computational purpose to the flat-Bessel replacement, but the map is local rather than a fixed turning-point rescaling: the effective radius and amplitude are determined by action matching at the requested χ. The Airy behaviour near the turning point is already built into the flat Bessel calculation. The separate second-order Airy formula discussed in Appendix A is a fallback approximation, not the definition of the main Olver-to-flat map.

This choice has two important consequences. First, the method is exact when $K = 0$. Second, most of the special-function complexity is delegated to the well-optimized flat Bessel implementation. The curved part of the problem becomes a coordinate map $z(\chi)$ and an amplitude factor $A(\chi)$.

5 The flat comparison equation

The reduced flat spherical-Bessel function is

$$v(z) = z j_l(\nu z). \tag{23}$$

It satisfies

$$v'' = \left[\frac{l(l+1)}{z^2} - \nu^2\right] v = \ell^2 \left[\frac{1}{z^2} - \alpha^2\right] v. \tag{24}$$

Compare this with the curved reduced equation (12). The only structural difference is

$$z \quad \text{versus} \quad S_K(\chi). \tag{25}$$

Therefore the leading-order Olver-to-flat approximation has the form

$$u_l^\nu(K,\chi) \simeq A(\chi) z(\chi) j_l(\nu z(\chi)), \tag{26}$$

with a closed-space sign factor inserted when $K = +1$. The unreduced function is

$$\phi_l^\nu(K,\chi) \simeq \frac{A(\chi)z(\chi)}{S_K(\chi)} j_l(\nu z(\chi)). \tag{27}$$

The implementation is literally this composition: compute z and A, call `BJL(l,nu*z,j_l)`, multiply by Az, and divide by S_K if the public result requested is ϕ rather than u.

6 The Olver action map

Define

$$f_K(\chi) = \alpha^2 - \frac{1}{S_K^2(\chi)}, \qquad f_0(z) = \alpha^2 - \frac{1}{z^2}. \tag{28}$$

The Olver coordinate is defined by equality of positive action distances from the turning point. For the curved equation define

$$Q_K(\chi) = \begin{cases} \int_\chi^{\chi_t} \sqrt{-f_K(s)}\, \mathrm{d}s, & \chi < \chi_t, \\ \int_{\chi_t}^{\chi} \sqrt{f_K(s)}\, \mathrm{d}s, & \chi > \chi_t, \end{cases} \tag{29}$$

with $Q_K(\chi_t) = 0$. In WKB language this is the accumulated action: below the turning point it measures the amount of exponential tunnelling, and above the turning point it measures the accumulated oscillatory phase. Points below the turning point map to $z < z_t$, and points above it map to $z > z_t$, where $z_t = 1/\alpha = \ell/\nu$.

The flat side is the same construction with f_0. It is convenient to remove the explicit α by setting $x = \alpha z$ and $y = \alpha r$. This defines the dimensionless flat action I_0:

$$I_0(x) = \begin{cases} \displaystyle\int_x^1 \sqrt{y^{-2} - 1}\,\mathrm{d}y = \log\left(\frac{1 + \sqrt{1 - x^2}}{x}\right) - \sqrt{1 - x^2}, & x < 1, \\ \displaystyle\int_1^x \sqrt{1 - y^{-2}}\,\mathrm{d}y = \sqrt{x^2 - 1} - \arccos\left(\frac{1}{x}\right), & x > 1, \end{cases} \tag{30}$$

with $I_0(1) = 0$. This definition is the flat side of (29): $x < 1$ is the evanescent side and $x > 1$ is the oscillatory side. The Olver map chooses z so that the flat comparison problem has the same action,

$$Q_K(\chi) = I_0(\alpha z), \qquad z_t = \frac{1}{\alpha} = \frac{\ell}{\nu}. \tag{31}$$

Why is the action the right thing to match? The intuition is the familiar WKB one from the Schrödinger equation. Away from the turning point the regular solution of $u'' = -\ell^2 f\, u$ behaves like $|f|^{-1/4} \exp(\pm \ell Q)$ on the evanescent side and $|f|^{-1/4} \cos(\ell Q - \frac{\pi}{4})$ on the oscillatory side, where ℓQ is the action integral (29). The action is the leading Liouville–Green invariant of the equation – the large, rapidly varying part of the solution – counting accumulated wavelengths above the turning point and tunnelling e-foldings below it. Two equations with different potentials f_K and f_0 but the same action measured from their turning points therefore have solutions that stay in step everywhere: same phase, same node count, same suppression. So by construction (31) the flat Bessel function $j_l(\nu z)$ at the mapped radius $z(\chi)$ already carries the curved phase and tunnelling. And because the turning point maps to the turning point ($Q_K = 0 \to I_0 = 0$), the construction is automatically smooth there, unlike a bare WKB amplitude $|f|^{-1/4}$.

For the curved cases the integral $Q_K(\chi)$ has analytic logarithm and atan2 forms, using $x = \alpha S_K(\chi)$. These are the same action integrals that appear in the older Langer/WKB formulae [4, 5]; CAMB uses them to define a map to the flat spherical-Bessel equation rather than to build an Airy argument directly. These closed forms are the expressions implemented in `qintegral_exact`; they avoid quadrature in a hot loop.

The remaining numerical task is to solve (31) for $x = \alpha z$. There is no simple elementary formula for the inverse of I_0, but each branch has a stable parameterization. Below the turning point write $x = \operatorname{sech} t$. Then

$$I_0(x) = t - \tanh t. \tag{32}$$

Above the turning point write $x = 1/\cos\theta$. Then

$$I_0(x) = \tan\theta - \theta. \tag{33}$$

Thus the code is not approximating the curved action at this stage; it is only inverting the exact flat comparison action. Near the turning point the action is small and both parameterizations have the same leading behaviour, $I_0 \simeq t^3/3$ or $I_0 \simeq \theta^3/3$. This is why `invert_flat_action` first forms

$$p = (3Q_K)^{1/3} \tag{34}$$

and uses a numerical polynomial fit in p^2 for x. These coefficients are not meant to be read as an analytic Taylor expansion of the Olver solution; they are fitted inverse-action formulae for the flat comparison problem in the narrow region where a direct root solve would be wasteful and where subtracting nearly equal logarithmic or trigonometric terms would lose precision. Away from the turning point the code switches to the natural asymptotic inverses: in the evanescent tail $t \simeq Q_K + 1$ and $x = \operatorname{sech} t$ is expanded in powers of $\exp[-(Q_K + 1)]$; on the oscillatory side θ is close to $\pi/2$, so the inverse is expanded in powers of $(Q_K + \pi/2)^{-1}$. This keeps the Olver call fast: the map requires the closed-form curved action, one branch choice, and a Horner polynomial or asymptotic formula before the final `BJL` call. Finally $z = x/\alpha = z_t x$, with the sign of the branch supplied by whether χ is below or above χ_t.

Matching the action fixes the phase but not the overall size; the amplitude follows by the same flux- (Wronskian-) conservation argument that gives the $|f|^{-1/4}$ prefactor of ordinary WKB. Differentiating the action-matching condition (31) gives the local Liouville map,

$$\frac{\mathrm{d}z}{\mathrm{d}\chi} = \sqrt{\frac{f_K(\chi)}{f_0(z)}}, \tag{35}$$

the ratio of the curved and flat local momenta. Writing the curved solution as the flat one carried through this map, $u(\chi) = A(\chi)\, z j_l(\nu z)$, the spurious first-derivative term cancels precisely when the amplitude is the Jacobian factor

$$A(\chi) = \left(\frac{\mathrm{d}z}{\mathrm{d}\chi}\right)^{-1/2} = \left(\frac{f_0(z)}{f_K(\chi)}\right)^{1/4} = \left|\frac{\alpha^2 - z^{-2}}{\alpha^2 - S_K(\chi)^{-2}}\right|^{1/4}, \tag{36}$$

which keeps the Wronskian constant. This Jacobian is not by itself the curved WKB amplitude: for $K = 0$, $z = \chi$ and $A = 1$. Rather, away from the turning point, if the flat comparison solution is also expanded in WKB form, its flat $|f_0|^{-1/4}$ amplitude multiplied by $A = (f_0/f_K)^{1/4}$ gives the curved $|f_K|^{-1/4}$ amplitude. At the turning point both numerator and denominator vanish, but the limit is finite. With $C_t = C_K(\chi_t)$,

$$z - z_t \simeq C_t^{1/3}(\chi - \chi_t), \qquad A(\chi_t) = C_t^{-1/6}. \tag{37}$$

For $K = 0$, $S_K = \chi$, $z = \chi$, and $A = 1$. Equations (26) and (27) then reduce exactly to the flat result.

7 Small-χ Olver expansion

Very near the origin the exact action map involves nearly equal quantities. The code therefore uses a local expansion of the differentiated map,

$$\left(\frac{\mathrm{d}z}{\mathrm{d}\chi}\right)^2 \left(\alpha^2 - \frac{1}{z^2}\right) = \alpha^2 - \frac{1}{S_K^2(\chi)}. \tag{38}$$

Write

$$z = \chi F, \qquad h = \frac{K}{\alpha^2} = K\frac{l(l+1)}{\nu^2}, \qquad a = K\chi^2. \tag{39}$$

The code's cubic-order expansion is

$$F = 1 - h\left[\frac{1}{6} + \frac{4a + 13h}{360} + \frac{48a^2 + 148ha + 737h^2}{45360}\right], \tag{40}$$

with

$$D = \frac{\mathrm{d}z}{\mathrm{d}\chi} = 1 - h\left[\frac{1}{6} + \frac{12a + 13h}{360} + \frac{240a^2 + 444ha + 737h^2}{45360}\right]. \tag{41}$$

The local approximation is then

$$\phi_l^\nu(\chi) \simeq \frac{\chi F}{S_K(\chi)} \frac{1}{\sqrt{D}}\, j_l(\nu\chi F). \tag{42}$$

In source-integration code this appears as an argument $x = \nu\chi F$ for the flat Bessel table and a weight proportional to $F/\sqrt{D}$, together with the geometrical factor $\chi/S_K(\chi)$.

8 Shifted-ν: the constant-map small-χ shortcut

Flat-Bessel replacements in the near-flat regime were also the central acceleration idea in the CLASS flat-rescaling approximation [6]. The shortcut here has the same computational purpose, but its constant rescaling is the constant part of CAMB's local Olver map rather than the CLASS turning-point rescaling γ with a fitted smooth amplitude.

The fastest source-integration shortcut is obtained by taking the local small-χ Olver map and keeping only its χ-independent part. In other words, the map is treated as a constant rescaling of the flat Bessel argument over the source interval. This keeps the computation as cheap as a flat table lookup, while still retaining the constant curvature corrections from the local Olver expansion.

Recall the small-χ variables

$$h = K\frac{l(l+1)}{\nu^2}, \qquad a = K\chi^2, \tag{43}$$

and write the local map as

$$F(\chi) = F_0 + F_2\chi^2 + F_4\chi^4 + \cdots, \qquad D(\chi) = \frac{\mathrm{d}z}{\mathrm{d}\chi} = D_0 + D_2\chi^2 + D_4\chi^4 + \cdots. \tag{44}$$

Setting $a = 0$ in (40) and (41) gives the constant terms

$$F_0 = D_0 = 1 - \frac{h}{6} - \frac{13h^2}{360} - \frac{737h^3}{45360}. \tag{45}$$

The shifted branch freezes the map at this value:

$$z \simeq F_0\chi, \qquad D \simeq F_0. \tag{46}$$

Substituting this into the local Olver expression (42) therefore gives

$$\phi_l^\nu(K,\chi) \simeq \frac{\chi F_0}{S_K(\chi)}\frac{1}{\sqrt{F_0}}\, j_l(\nu F_0\chi) = \sqrt{F_0}\,\frac{\chi}{S_K(\chi)} j_l(\nu_{\mathrm{eff}}\chi), \qquad \nu_{\mathrm{eff}} = \nu F_0. \tag{47}$$

For the reduced function this is equivalently

$$u_l^\nu(K,\chi) \simeq \sqrt{F_0}\,\chi j_l(\nu_{\mathrm{eff}}\chi). \tag{48}$$

This is exactly what the production code implements in the shifted near-flat paths:

$$\texttt{nu\_eff} = \nu\,\texttt{F0}, \qquad \texttt{nu\_amp} = \sqrt{\texttt{F0}}, \tag{49}$$

followed by a lookup of the already-tabulated flat j_l at $\nu_{\mathrm{eff}}\chi$, multiplication by `nu_amp`, and the geometrical factor $\chi/S_K(\chi)$ for the unreduced value.

It is useful to expand the effective wavenumber squared:

$$\nu_{\mathrm{eff}}^2 = \nu^2 F_0^2 = \nu^2\left[1 - \frac{h}{3} - \frac{2h^2}{45} - \frac{58h^3}{2835} + \mathcal{O}(h^4)\right]. \tag{50}$$

The first correction in (50) is $-\nu^2 h/3 = -Kl(l+1)/3$. This is the same leading constant curvature correction obtained by expanding $1/S_K^2$ in the radial equation, as shown in the note below. The implemented F_0 form also keeps the constant h^2 and h^3 corrections from the Olver map at essentially no extra per-sample cost.

The shifted-ν gate is tied to the part of the local Olver map that this constant shortcut omits. With

$$t = \alpha\chi, \qquad \alpha = \frac{\nu}{\ell}, \qquad \ell = \sqrt{l(l+1)}, \tag{51}$$

the first omitted coordinate term is

$$F(\chi) - F_0 = -\frac{ha}{90} + \cdots = -\frac{t^2}{90\alpha^4} + \cdots, \tag{52}$$

where the second equality uses $K^2 = 1$, which holds on every non-flat ($K = \pm 1$) code path that takes this branch, so that $ha = K^2\chi^2/\alpha^2 = \chi^2/\alpha^2 = t^2/\alpha^4$. The leading omitted flat-Bessel argument shift is then

$$|\delta x| = |\nu\chi\,[F(\chi) - F_0]| \simeq \frac{\ell t^3}{90\alpha^4}. \tag{53}$$

The leading omitted amplitude correction follows from $F/\sqrt{D}$ relative to $\sqrt{F_0}$:

$$\frac{F/\sqrt{D}}{\sqrt{F_0}} - 1 \simeq \frac{t^2}{180\alpha^4}. \tag{54}$$

At the endpoint $t_{\max} = \alpha\chi_{\max}$, the current code combines these as

$$\epsilon_{\text{shift}} = \frac{1}{2}\frac{\ell\, t_{\max}^3}{90\alpha^4} + \frac{t_{\max}^2}{180\alpha^4}, \tag{55}$$

with a threshold scaled by the requested accuracy. This shifted-ν test is not the whole source-range decision: the caller first requires the common near-flat range gate (71). Thus the shifted-ν branch is used only when the local small-χ Olver map is valid across the range and freezing that map to F_0 is accurate enough. The error estimate above is a comparison against omitted local Olver terms, rather than a pointwise relative-error test, which would be inappropriate near Bessel zeros.

Note: a simpler constant-potential shift that is not used

There is a nearby approximation that looks slightly simpler algebraically, and it is useful for understanding the leading term. Expand the curvature centrifugal factor directly:

$$\frac{1}{S_K^2(\chi)} = \frac{1}{\chi^2} + \frac{K}{3} + \frac{K^2\chi^2}{15} + \frac{2K^3\chi^4}{189} + \cdots. \tag{56}$$

Substituting this into the reduced oscillator equation gives

$$u'' + \left[\nu^2 - \frac{L}{S_K^2(\chi)}\right] u = 0, \qquad L = l(l+1), \tag{57}$$

$$u'' + \left[\nu^2 - \frac{L}{\chi^2} - \frac{KL}{3} + \mathcal{O}(K^2 L\chi^2)\right] u = 0. \tag{58}$$

If only the constant curvature correction is retained, the result is a flat reduced spherical-Bessel equation with

$$\nu_{\text{pot}}^2 = \nu^2 - \frac{Kl(l+1)}{3} = \nu^2\left(1 - \frac{h}{3}\right). \tag{59}$$

The corresponding amplitude-corrected form would be

$$\phi_l^\nu(\chi) \approx \left(\frac{\nu_{\text{pot}}}{\nu}\right)^{1/2} \frac{\chi}{S_K(\chi)} j_l(\nu_{\text{pot}}\chi). \tag{60}$$

This agrees with the implemented formula through first order in h, since

$$\frac{\nu_{\text{pot}}}{\nu} = \sqrt{1 - \frac{h}{3}} = 1 - \frac{h}{6} - \frac{h^2}{72} + \cdots, \tag{61}$$

whereas

$$F_0 = 1 - \frac{h}{6} - \frac{13h^2}{360} + \cdots. \tag{62}$$

The constant-potential version is therefore a legitimate additional approximation, but it is not the one used in the current code: it has similar numerical cost and drops constant higher-order Olver information that is already available in F_0.

9 Accuracy gates and approximation hierarchy

The formulas above are used as a controlled hierarchy, not as a single unconditional replacement for the exact hyperspherical Bessel function. It is helpful to separate the pointwise Bessel call from the source-integration fillers.

9.1 What the full action-map Olver error estimate is based on

The "full action-map Olver" path in the code means the full analytic action map $z(\chi)$ and amplitude $A(\chi)$, but it is still the leading-order uniform Olver approximation. Higher Olver terms are not evaluated. A useful way to see the first omitted term is to substitute

$$u(\chi) = A(\chi)w(z(\chi)), \qquad A = (\mathrm{d}z/\mathrm{d}\chi)^{-1/2},$$

back into the exact reduced equation. The comparison function w then obeys

$$\frac{\mathrm{d}^2 w}{\mathrm{d}z^2} + \left[\ell^2 f_0(z) + \Psi(z)\right] w = 0, \qquad \Psi(z) = -\frac{1}{2(\mathrm{d}z/\mathrm{d}\chi)^2}\{z,\chi\}_S, \tag{63}$$

where

$$\{z,\chi\}_S = \frac{z'''}{z'} - \frac{3}{2}\left(\frac{z''}{z'}\right)^2 \tag{64}$$

is the Schwarzian derivative. The leading approximation drops Ψ and solves only the flat comparison equation. The next Olver correction is driven by this residual term, or equivalently by its integral against the flat comparison solutions.

This gives the form of the gates, but not trustworthy constants by itself. In the near-flat regime the local expansion of the omitted curvature terms is controlled by powers of $h = K/\alpha^2$; after the leading action and amplitude matching, the relevant curvature-dependent remainder falls rapidly with α, with the observed raw-Olver envelope consistent with an α^{-4} scaling in the high-l open-space regime. This motivates an α-based fast-path guard there. At low l, in open cases with very small ν/l, and in closed-space endpoint tails, the validation errors are better organized by the endpoint metrics $\chi/(2\nu)$ in open models and $\chi/[2(\nu-l)]$ in closed models. The code therefore combines a high-l open-space α shortcut with metric gates, and otherwise falls back to stable recurrence. The numerical thresholds are calibrated on peak-normalized errors; a local relative error or the pointwise ratio $\Psi/(\ell^2 f_0)$ would be misleading near turning points and Bessel zeros.

9.2 Pointwise Bessel calls

The full action-map Olver value means the `compute_olver_z_amp` path: compute z and A, call `BJL(l,nu*z,j_l)`, and multiply by the transport amplitude. This leading-order value is exact in the flat limit, but for $K = \pm 1$ it is still an asymptotic approximation. The public pointwise calls therefore apply empirical gates before accepting that raw full action-map Olver value.

The decision order in `olver_reduced` is:

1. For $l \le 2$, use `phi_recurs`.

2. For $l > 2$, first try the local small-χ approximation to the Olver map, if

$$\left[\frac{\nu}{l} > 2.5 \quad \text{or} \quad \left(l \ge 50 \text{ and } \frac{\nu}{l} > 1\right)\right], \qquad \frac{l^2\chi^7}{\nu} < 5\times 10^{-2}. \tag{65}$$

   This pointwise gate deliberately uses ν/l and l^2 as cheap high-l proxies; the map itself still uses the more accurate $\sqrt{l(l+1)}$ curvature scale.

3. If the small-χ map is not used, the open model accepts the full action-map Olver value without an additional endpoint metric restriction only when ν/l clears the calibrated continuous cutoff

$$\frac{\nu}{l} \geq \alpha_{\text{open}}(l), \qquad \alpha_{\text{open}}(l) = \max\left[0.095,\, 0.12\left(\frac{500}{l}\right)^{p(l)}\right], \tag{66}$$

with $p(l) = 0.70$ for $l < 500$ and $p(l) = 0.14$ for $l \geq 500$. Raw open-space cutoff scans give $\alpha_{\text{open}} \simeq 0.12$ near $l = 500$ and $\alpha_{\text{open}} \simeq 0.095$ by $l \simeq 2000$, remaining there through $l = 10000$. Outside that regime, open models accept the full action-map value only in the small endpoint corner

$$\frac{\chi}{2\nu} \leq 3.0 \times 10^{-3}. \tag{67}$$

4. In closed models, accept the full action-map Olver value only where the finite-spectrum endpoint metric is below threshold:

$$\frac{\chi}{2(\nu - l)} \leq 7.0 \times 10^{-3}. \tag{68}$$

Otherwise use the fallback sequence described below. This closed-space metric is still applied at large ν/l: low-l and endpoint cases showed that an unconditional $\nu/l \geq 3$ pointwise shortcut is too broad.

Here "fallback" does not necessarily mean immediate recurrence. At high l, the code first tries the second-order Airy one-point approximation derived in Appendix A: $l \geq 100$ in open models and $l \geq 150$ in closed models, subject to the Airy-specific gates in that appendix. If this Airy branch is outside its calibrated range, open models then have a narrow small-ν Macdonald-function fallback (Appendix B) before the exact recurrence `phi_recurs`. This branch is rarely reached in ordinary runs, because the Airy fallback covers most high-l open points where it would otherwise apply.

For closed models this fallback sequence is not a large uncontrolled slow region. The argument has already been folded into $0 \leq \chi \leq \pi/2$, so failing the closed leading-Olver gate implies

$$\nu - l < \frac{\chi}{2(7.0 \times 10^{-3})} \leq 112.3.$$

Inside this region, the closed Airy fallback has the additional gate

$$\lambda(\beta^2 - 1) = \frac{\nu^2 - \lambda^2}{\lambda} = \frac{(l + d)^2 - (l + 1/2)^2}{l + 1/2} \geq 15, \qquad d = \nu - l.$$

For large l this is approximately $2d - 1 \geq 15$, so the Airy branch covers roughly $d \gtrsim 8$. High-l closed points that miss both the leading Olver and Airy gates are therefore very close to the finite spectral endpoint, with small Gegenbauer degree $n = d - 1$. For $l < 150$, direct recurrence is cheap because the requested order itself is modest. In both cases the finite-spectrum endpoint or Gegenbauer start keeps the exact fallback cheap enough for the occasional seed and re-seed values used by the source integrator. These thresholds are calibrated on peak-normalized Bessel error on a validation grid that includes dense low-χ points, turning-point stencils, closed endpoints, $l \leq 10000$, and open-space tails about eighty oscillations past the turning region. The strict envelope is kept close to 10^{-4}; a small number of small-χ fast-path points can sit between 10^{-4} and 2×10^{-4}, which is accepted to retain the faster local map.

9.3 Source-integration gates

In the line-of-sight source integration, `u_olver` is usually not called at every source time. It supplies starting and re-seeding values for Numerov stepping, while neighbouring values are advanced with the differential equation. As above, $B_{\rm acc}$ denotes the accuracy-boost factor used for non-flat source integration, bounded below by one. The Numerov step size and re-bootstrap interval are scaled by $B_{\rm acc}$.

Before Numerov, near-flat scalar source ranges can be filled directly from flat Bessel tables. These shortcuts are also tied to a peak-position sampling ratio, defined via the sampling proxy

$$\vartheta_{\rm samp} = \frac{r_s(z_{\rm drag})}{D_M(\tau_0)}, \qquad \vartheta_{\rm samp}^{\rm P18} = \left[\frac{r_s(z_{\rm drag})}{D_M(\tau_0)}\right]_{\rm Planck\,2018}, \tag{69}$$

where $D_M(\tau_0)$ is the comoving angular-diameter distance corresponding to the full present conformal time τ_0. Although the numerator uses $r_s(z_{\rm drag})$, $\vartheta_{\rm samp}$ is deliberately not the usual drag or acoustic angular scale: the standard quantity divides the sound horizon by the comoving angular-diameter distance *to the drag (or last-scattering) epoch*, whereas here the denominator is the distance over the *total* present conformal time. It is used only as a cheap, close approximate proxy for the acoustic peak positions in the sampling-ratio gate below. The sampling ratio is

$$s_{\rm peak} = \frac{\vartheta_{\rm samp}^{\rm P18}}{\vartheta_{\rm samp}}. \tag{70}$$

The gate is one-sided, matching the current l-sampling logic. When $s_{\rm peak} \geq 1 - 0.03$, CAMB keeps the fiducial l-sampling strategy and can reuse the flat spherical-Bessel table machinery, enlarging the maximum tabulated argument when needed for shifted-ν calls. If $s_{\rm peak} > 1$, the acoustic peaks are wider-spaced in l than in the fiducial model, but the code does not coarsen the sampled l grid, so the existing Bessel-table sampling remains conservative. Only when $s_{\rm peak} < 1 - 0.03$ are the peaks compressed enough that the l-sampling is made denser; in that case these near-flat table shortcuts are not assumed. This avoids recomputing a different Bessel sampling for small geometric rescalings while keeping the compressed-peak case safe. In ordinary CMB parameter fits this bookkeeping gate almost always passes: models that fit the acoustic peaks have θ_* measured very accurately, and this proxy tracks the same peak-position shift closely enough to keep $s_{\rm peak}$ near unity. The gate mainly protects exploratory geometries whose peak scale is already far from the data-supported value.

Passing this bookkeeping gate does not by itself choose a Bessel approximation. It only allows CAMB to consider the two source-integration branches that reuse the flat spherical-Bessel table: the shifted-ν lookup and the local small-χ Olver-map lookup. Both of these branches must then pass their Bessel-accuracy gates. They first require the common near-flat range gate

$$\left[\frac{\nu}{l} > 2.5 \quad \text{or} \quad \left(l \geq 50 \text{ and } \frac{\nu}{l} > 1\right)\right], \qquad \frac{l^2 \chi_{\rm max}^7}{\nu} < \frac{0.1}{B_{\rm acc}}. \tag{71}$$

This gate uses ν/l and l^2 as cheap high-l proxies for the curvature scale in the hot source-integration path. Relative to the earlier source gate, it is stricter in the endpoint metric but less restrictive in ν/l, which keeps the tested high-l, low-ν/l near-flat band.

1. If the common range gate passes, `UseSmallChiNearFlatIntegration` may fill the range using $j_l(\nu\chi F)$ and the $F/\sqrt{D}$ weight.

2. The shifted-ν filler uses the same common range gate, and then adds the constant-map residual test from (55):

$$\epsilon_{\rm shift} < \frac{10^{-3}}{B_{\rm acc}}, \tag{72}$$

   with $\epsilon_{\rm shift}$ defined in (55). Thus shifted-ν is used only when the constant-map Olver limit is valid and the constant-map approximation to that map is valid.

The pointwise `phi_olver` small-χ fast path remains separate and keeps the tighter local threshold $l^2\chi^7/\nu < 5 \times 10^{-2}$, because it must preserve the pointwise peak-normalized Bessel envelope at individual χ values rather than integrated source weights.

If neither flat-table filler is used, the code follows the general non-flat source-integration route. A pointwise Bessel evaluator supplies seed and re-seed values, usually from the Olver-to-flat approximation, but with the same high-l Airy and recurrence fallbacks selected by the pointwise gates. Numerov then propagates these seeds to nearby source samples. Thus `phi_recurs` is reached through the pointwise hierarchy, not as an automatic consequence of rejecting the table fillers. The overall source accuracy is therefore controlled jointly by the pointwise Olver gates, the near-flat filler gates, and the Numerov sampling, not by the raw Olver asymptotic formula alone.

10 Stable recursive fallback: exact seeds, upward recurrence and Miller recurrence

The action-map Olver path is a high-throughput leading-order uniform approximation. For low l, invalid asymptotic corners, and some fallback regions, CAMB uses direct recurrence through `phi_recurs`. This recurrence machinery follows the standard hyperspherical-Bessel relations and the later stable-recursion improvements developed for CLASS [3, 5, 6].[1] This section explains that routine because it is part of the actual new path.

The exact seed functions are

$$\phi_0^\nu(K,\chi) = \frac{\sin(\nu\chi)}{\nu S_K(\chi)}, \tag{73}$$

with Taylor safeguards for small arguments, and

$$\phi_1^\nu(K,\chi) = \frac{\cot_K \chi \, \sin(\nu\chi)/\nu - \cos(\nu\chi)}{S_K(\chi) b_1(K,\nu)}. \tag{74}$$

For $K = 0$, these reduce to $j_0(\nu\chi)$ and $j_1(\nu\chi)$.

The exact recurrence is

$$b_j \phi_j^\nu = (2j-1)\cot_K \chi \, \phi_{j-1}^\nu - b_{j-1}\phi_{j-2}^\nu, \qquad j \geq 2. \tag{75}$$

where $b_j = b_j(K,\nu) = \sqrt{\nu^2 - Kj^2}$. Forward recurrence from $j = 0, 1$ is used only where it is stable, essentially the safely oscillatory region. Away from the open-space refinements described below, the code tests a condition of the form

$$b_l > 0, \qquad |\cot_K \chi| < \frac{b_l}{\max(1,l)}. \tag{76}$$

For $K = -1$ this comparison is made in the algebraically equivalent variable

$$y_{\rm open} = \frac{\nu \sinh\chi}{l}, \qquad \coth\chi < \frac{\sqrt{\nu^2 + l^2}}{l} \quad \Longleftrightarrow \quad y_{\rm open} > 1. \tag{77}$$

[1] A direct comparison with CLASS v3.3.4 shows the same seed formulae, upward recurrence, Miller downward recurrence, continued-fraction logarithmic derivative, and closed-space Gegenbauer start. The main implementation differences are that CLASS builds interpolation tables over l and χ, while CAMB's `phi_recurs` is a pointwise reference routine; CLASS uses its `get_CF1` continued-fraction helper for some closed-space Miller starts before falling back to Gegenbauer, while CAMB either starts closed-space Miller recursion from the finite endpoint condition $b_\nu\phi_\nu = 0$ or, near the finite-spectrum endpoint, uses G and G' from the Gegenbauer recurrence to set the Miller pair directly; CLASS normalizes the Miller sequence with ϕ_0, while CAMB normalizes with whichever of ϕ_0 or ϕ_1 is numerically safer; and CAMB computes G and G' together in the Gegenbauer recurrence rather than using the final derivative identity with a division by $1 - x^2$. Here `get_CF1` is the CLASS routine evaluating the modified continued fraction for $D_l = (\phi_l^\nu)'/\phi_l^\nu$, with an auxiliary sign used to choose the arbitrary Miller start amplitude.

This avoids comparing two quantities that are both very close to one when $\nu/l \ll 1$. Since the continued fraction used to start Miller recurrence can be slow just below the open turning boundary, CAMB also uses forward recurrence for

$$y_{\text{open}} \geq 1 - 5 \times 10^{-3}. \tag{78}$$

Peak-normalized forward/Miller scans show that the very small-ν/l family has a wider safe forward region, so CAMB also uses

$$\frac{\nu}{l} \leq 2.1 \times 10^{-3}, \qquad y_{\text{open}} \geq 0.9. \tag{79}$$

Tiny-χ, small-ν/l cases such as $l = 4, \nu/l = 0.002$ remain on Miller recurrence.

Outside the stable forward region, forward recurrence would amplify the unwanted independent solution. The code then uses Miller downward recurrence,

$$\phi^{\nu}_{j-1} = \frac{(2j+1)\cot_K \chi\, \phi^{\nu}_j - b_{j+1}\phi^{\nu}_{j+1}}{b_j}, \qquad j = l, l-1, \ldots, 1. \tag{80}$$

Downward recurrence gives the correct ratios but needs a top boundary condition. For flat and open models this is supplied by a logarithmic derivative. For closed models CAMB can also use the finite endpoint of the discrete spectrum.

10.1 The logarithmic derivative used to start downward recurrence

The derivative identity is

$$\frac{\mathrm{d}\phi^{\nu}_j}{\mathrm{d}\chi} = j \cot_K \chi\, \phi^{\nu}_j - b_{j+1}\phi^{\nu}_{j+1}. \tag{81}$$

Define

$$D_j = \frac{(\phi^{\nu}_j)'}{\phi^{\nu}_j}. \tag{82}$$

Then

$$b_{j+1}\frac{\phi^{\nu}_{j+1}}{\phi^{\nu}_j} = j \cot_K \chi - D_j. \tag{83}$$

For the logarithmic-derivative start, set an arbitrary $\widetilde{\phi}_l = 1$, compute D_l, set

$$b_{l+1}\widetilde{\phi}_{l+1} = l \cot_K \chi - D_l, \tag{84}$$

then recur downward with (80). At the end the arbitrary sequence is normalized by matching the exact seed values (73) or (74), whichever is numerically safer.

For $K = +1$, ν is an integer and the regular closed spectrum ends at $l = \nu - 1$. The recurrence coefficient is then $b_j(+1, \nu) = \sqrt{\nu^2 - j^2}$, so $b_\nu = 0$. CAMB can choose an arbitrary $\widetilde{\phi}_{\nu-1}$, set the endpoint product $b_\nu \widetilde{\phi}_\nu$ to zero, and run the same Miller recurrence downward to the desired l and then to the exact low-order seeds. The code uses this finite-endpoint start when $\nu - l > 64$, equivalently when the Gegenbauer degree $n = \nu - l - 1$ is at least 64. For smaller n, the Gegenbauer start below is cheaper and is used directly.

For $K = 0$ and $K = -1$, D_l is obtained from a continued fraction. Let

$$R_j = b_{j+1}\frac{\phi_{j+1}}{\phi_j}. \tag{85}$$

The recurrence gives

$$R_j = \frac{b^2_{j+1}}{(2j+3)\cot_K \chi - R_{j+1}}, \tag{86}$$

so

$$D_l = l \cot_K \chi - \cfrac{b_{l+1}^2}{(2l+3)\cot_K \chi - \cfrac{b_{l+2}^2}{(2l+5)\cot_K \chi - \cdots}}. \quad (87)$$

The implementation evaluates this with a modified continued-fraction iteration.

10.2 Closed-space Gegenbauer Miller start

For $K = +1$, the regular solution has an exact finite Gegenbauer-polynomial form,

$$\phi_l^\nu(+1, \chi) = N_{l\nu} \sin^l \chi \, C_{\nu-l-1}^{l+1}(\cos \chi). \quad (88)$$

This identity is the key observation in Tram's closed-space stable-recursion algorithm, and Lesgourgues–Tram use it to remove the old obstruction to closed-space backward recurrence [5, 6]. Let

$$n = \nu - l - 1, \qquad \lambda = l + 1, \qquad x = \cos \chi, \qquad G(x) = C_n^\lambda(x). \quad (89)$$

The corresponding logarithmic derivative would be

$$D_l = l \cot \chi - \sin \chi \, \frac{G'(\cos \chi)}{G(\cos \chi)}. \quad (90)$$

However the Miller recurrence does not need D_l itself. It needs the starting pair $\widetilde{\phi}_l$ and $b_{l+1}\widetilde{\phi}_{l+1}$. CAMB therefore first forms the algebraically equivalent pair

$$\widetilde{\phi}_l = G(\cos \chi), \qquad b_{l+1}\widetilde{\phi}_{l+1} = \sin \chi \, G'(\cos \chi). \quad (91)$$

When G is safely non-zero, the arbitrary Miller amplitude is rescaled to $\widetilde{\phi}_l = 1$ and $b_{l+1}\widetilde{\phi}_{l+1} = \sin \chi \, G'/G$; near a zero of G, CAMB keeps the unscaled pair in (91). This prevents a zero of ϕ_l^ν from becoming a log-derivative failure, while still using the cheapest scaled start away from zeros. It also avoids cancellation between $l \cot \chi$ and D_l near the closed poles. This is the closed-case `phi_closed_gegenbauer_start`. The code computes G and G' together by the Gegenbauer recurrence

$$C_0^\lambda(x) = 1, \qquad C_1^\lambda(x) = 2\lambda x, \quad (92)$$

$$C_k^\lambda(x) = \frac{2(k + \lambda - 1)x C_{k-1}^\lambda(x) - (k + 2\lambda - 2)C_{k-2}^\lambda(x)}{k}, \quad (93)$$

with the derivative recurrence obtained by differentiating this relation. This provides a stable top boundary condition for the Miller recurrence in a finite closed spectrum. The Gegenbauer calculation is not used as a separate final approximation; it only supplies the Miller start in (91).

11 How CAMB actually uses the approximations in source integration

The line-of-sight source integral needs the Bessel function at many neighbouring time points. The computational pressure is the same one discussed in the non-flat CLASS implementation: hyperspherical Bessel functions are too expensive to store globally and too common in the transfer integral to evaluate with a slow special-function path [6]. CAMB's main path therefore uses a hierarchy rather than calling `u_olver` independently at every source time.

For flat models, `DoFlatIntegration` uses precomputed flat Bessel tables. It converts each time point to a flat argument $x = q(\tau_0 - \tau)$, finds the bracketing table point, and evaluates a cubic spline in Horner form. This is the baseline fast method.

For near-flat models there are two table-based shortcuts. `DoNearFlatIntegration` uses the shifted-ν argument $x = \nu_{\rm eff}\chi$, with

$$h = K\frac{l(l+1)}{\nu^2}, \qquad F_0 = 1 - \frac{h}{6} - \frac{13h^2}{360} - \frac{737h^3}{45360}, \qquad \nu_{\rm eff} = \nu F_0, \qquad \nu_{\rm amp} = \sqrt{F_0}. \tag{94}$$

The tabulated j_l value is multiplied by this amplitude factor and by precomputed time/geometric weights. `DoNearFlatSmallChiIntegration` instead uses the small-χ Olver map $x = \nu\chi F$ and weights proportional to $F/\sqrt{D}$. Both paths reduce the curved Bessel problem to looking up the existing flat Bessel table.

For genuinely non-flat source integration, `IntegrateSourcesBessels` first chooses a starting point near the turning point,

$$\chi_{\rm diss} = S_K^{-1}(\ell/\nu), \tag{95}$$

then evaluates u_l^ν there using `u_olver`. The source integral is split into ranges above and below this starting point. On each range, `DoRangeInt` either fills all grid values with a near-flat approximation, or steps the reduced differential equation by Numerov integration.

12 Nearby values by Numerov integration

Numerov integration is used because the reduced equation has the special form

$$y'' = Q(\chi)y, \qquad Q(\chi) = \frac{l(l+1)}{S_K^2(\chi)} - \nu^2, \tag{96}$$

with no first-derivative term. This is precisely the form for which Numerov's method is efficient and high-order accurate.

Let the integration step (the signed substep $\Delta\chi_{\rm sub}$) be denoted η; it is distinct from the curvature parameter h of the Olver sections above. Write $Q_n = Q(\chi_n)$ and $y_n = y(\chi_n)$. The Numerov update used in the code is

$$y_{n+1} = \frac{2\left(1 + \frac{5\eta^2 Q_n}{12}\right) y_n - \left(1 - \frac{\eta^2 Q_{n-1}}{12}\right) y_{n-1}}{1 - \frac{\eta^2 Q_{n+1}}{12}}. \tag{97}$$

This is a fourth-order local correction to the ordinary three-point second-derivative formula. Because (97) is a three-point (two-step) recurrence, it needs *two* consecutive accurate values to start; once these are known from `u_olver`, the neighbouring values satisfy the same smooth ODE.

The code initializes as follows. At the start of a range it takes y_0 from the end of the previous range if one is available, and otherwise evaluates

$$y_0 = u_l^\nu(K, \chi_0) \tag{98}$$

using `u_olver`. For the very first substep it then evaluates the next point $y_1 = u_l^\nu(K, \chi_0 + \Delta\chi_{\rm sub})$ with a second `u_olver` call; these two values are what start the Numerov recurrence. After that, the update (97) propagates y to all intermediate substeps. At each source grid point the stored value is

$$\phi_l^\nu(\chi_n) = \frac{y_n}{S_K(\chi_n)}. \tag{99}$$

Two implementation details keep this loop both cheap and stable. First, the metric factor $S_K(\chi)$ (and $C_K(\chi)$, needed to advance it) is not recomputed with a fresh `sin`/`sinh` call at every substep. Instead the code steps S_K, C_K forward by the fixed substep $\Delta\chi_{\rm sub}$ using the angle-addition formulae, which for the constant step reduce to a two-term linear update with precomputed coefficients $S_K(\Delta\chi_{\rm sub}), C_K(\Delta\chi_{\rm sub})$:

$$\begin{aligned} K=+1: \quad & \begin{cases} s_{n+1} = s_n \cos\Delta\chi_{\rm sub} + c_n \sin\Delta\chi_{\rm sub}, \\ c_{n+1} = c_n \cos\Delta\chi_{\rm sub} - s_n \sin\Delta\chi_{\rm sub}, \end{cases} \\ K=-1: \quad & \begin{cases} s_{n+1} = s_n \cosh\Delta\chi_{\rm sub} + c_n \sinh\Delta\chi_{\rm sub}, \\ c_{n+1} = c_n \cosh\Delta\chi_{\rm sub} + s_n \sinh\Delta\chi_{\rm sub}, \end{cases} \end{aligned} \tag{100}$$

with $s_n = S_K(\chi_n)$, $c_n = C_K(\chi_n)$. Then Q_n follows from s_n alone. Second, both the angle-addition recursion and the Numerov recursion accumulate floating-point drift over many steps, so the code periodically re-seeds: it recomputes S_K, C_K directly from χ, and re-evaluates the two Numerov starting values with fresh `u_olver` calls before continuing. The default re-bootstrap interval is 200 substeps, reduced by the non-flat integration accuracy boost. A shorter 25-substep interval is used for very low effective multipoles, $l/s_{\rm peak} < 25$, and for the moderate-ν, high-ν/l band

$$\nu < 1000, \qquad \frac{\nu}{l} > 2, \tag{101}$$

where low- and mid-L EE is sensitive to small Numerov phase drift. This nu-aware gate avoids applying the frequent re-seeding to all low l modes, while correcting the modes where source integration has too few weighted oscillations to average the drift away. The minimum interval after accuracy scaling is 25 substeps. This bounds the accumulated error without paying for an `u_olver` call at every step.

12.1 Direct Olver evaluation for high-substep ranges

The Numerov step $\eta = \Delta\chi_{\rm sub}$ is capped at $\Delta\chi_{\rm max} \simeq 0.3/\nu$ (the precise coefficient depends on ℓ and is scaled by $B_{\rm acc}$), independent of the source-grid spacing $\Delta\chi_{\rm source}$. In genuinely non-flat models $\Delta\chi_{\rm source}$ can be several times $\Delta\chi_{\rm max}$, so a single source-grid step then requires

$$n_{\rm sub} = \max\left(1, \left\lceil \frac{\Delta\chi_{\rm source}}{\Delta\chi_{\rm max}} \right\rceil\right) \tag{102}$$

Numerov substeps. Profiling shows one `u_olver`/`phi_olver` evaluation costs about the same as 13 Numerov substeps, while closed models spend most of their substeps in ranges with $n_{\rm sub}$ of order ten to a few tens. When $n_{\rm sub} \geq n_\star = 10$, `DoRangeInt` therefore fills ϕ_l^ν at every source-grid point directly from `phi_olver`, in `FillDirectOlverPhiVals`, rather than Numerov-stepping the finer $\Delta\chi_{\rm max}$ grid between them. This is a strict accuracy improvement as well as a speed-up.

The source integral is then accumulated by a trapezoidal rule: the first and last stored Bessel values in the range get half weight, and the final sum is multiplied by $\Delta\chi/\sqrt{|\mathcal{K}|}$, equivalently the conformal-time step.

This explains why the Olver approximation is used differently inside source integration than in a standalone point call. It supplies a robust starting value near the peak or turning point. Nearby values are cheaper and sufficiently accurate from the differential equation itself. The step size is limited to a fraction of $1/\nu$, with accuracy-boost factors, so that the oscillation is well sampled. The integration stops early in dissipative tails when $|\phi_l^\nu|$ is below a threshold or when the solution is clearly moving into the exponentially small closed-space tail.

13 Near-flat grid fillers inside `DoRangeInt`

Before doing Numerov, `DoRangeInt` checks two fast fillers:

1. `UseShiftedQScalarApprox`: if the model is sufficiently near flat and the shifted-ν error estimate passes, fill all source-grid Bessel values directly from $j_l(\nu F_0 \chi)$ with the amplitude $\sqrt{F_0}$ and the χ/S_K geometrical factor.

2. `UseSmallChiNearFlatIntegration`: if the small-χ Olver map is accurate on the whole range, fill all values from $j_l(\nu\chi F)$ with the $F/\sqrt{D}$ amplitude and χ/S_K geometrical factor.

These branches avoid stepping the full hyperspherical ODE at all. They are particularly useful in nearly flat models, where the exact curved function is very close to a flat Bessel function but a discontinuous switch between exact flat and non-flat treatment would be numerically undesirable. This is the same continuity goal emphasized for the CLASS flat-rescaling approximation, although the CAMB fillers use the local Olver coefficients F, D and F_0 rather than the CLASS γ, A^l_ν model [6].

A useful way to think about the hierarchy is:

Regime	Computation	Reason
Flat	table/spline of $j_l(q\chi)$	exact flat result.
Near-flat, very small curvature effect	constant-map shifted-ν table lookup	fastest limit of small-χ Olver.
Near-flat, local small-χ range	table lookup at $\nu\chi F$, weight $F/\sqrt{D}$	retains next curvature terms in Olver map.
General non-flat range	start with `u_olver`, propagate by Numerov	avoids expensive repeated Olver calls.
Non-flat range needing ≥ 10 Numerov substeps per source step	direct `phi_olver` at each source-grid point (`FillDirectOlverPhiVals`)	one `phi_olver` call ≈ 13 substeps; removes accumulated Numerov phase drift.
High-l pointwise fallback	second-order Airy one-point approximation	cheaper than recurrence in its calibrated gates; derived in Appendix A.
Open post-Airy, very small ν	action-matched scaled $K_{i\nu}$	rare fallback for $K = -1$, described in Appendix B.
Low-order or post-fallback remainder	`phi_recurs`	exact recurrence is safer; high-l closed modes left after Airy have very small $n = \nu - l - 1$.

14 Closed-space symmetry handling

For $K = +1$, both the Olver path and `phi_recurs` fold χ into $[0, \pi/2]$. The signs come from the exact closed-space representation (88) and the closed-space parity relations described in the earlier hyperspherical-Bessel literature [4, 5]. Reducing χ modulo 2π, reflecting around π, and reflecting around $\pi/2$ introduces factors $(-1)^l$ and $(-1)^{\nu-l-1}$. The accumulated factor is called `symm`. The final reduced result is therefore really

$$u^\nu_l(+1, \chi) \simeq \texttt{symm}\, A z j_l(\nu z). \tag{103}$$

It is only the `symm` factor that is being claimed exact here: the folding into $[0, \pi/2]$ and its sign are an exact parity property of the closed eigenfunctions, not an approximation. The remaining $Azj_l(\nu z)$ is still the leading-order Olver approximation, as the $\simeq$ indicates; folding changes the sign, not the accuracy.

15 How all pieces fit together

For a single standalone call in the main Olver region, the compact formula is

$$\boxed{u^\nu_l(K, \chi) \simeq \sigma A(\chi) z(\chi) j_l(\nu z(\chi))} \tag{104}$$

with $\sigma = 1$ for $K = 0, -1$ and $\sigma = \texttt{symm}$ for $K = +1$. The public unreduced function is

$$\boxed{\phi_l^\nu(K,\chi) \simeq \frac{\sigma A(\chi) z(\chi)}{S_K(\chi)} j_l(\nu z(\chi)).} \tag{105}$$

For many neighbouring source points, CAMB usually does not repeat this full calculation: table fillers or Numerov stepping take over as described above.

16 Where the Airy approximation fits

The distinction in Sec. 4 is the main one: CAMB's default Olver path uses a flat spherical Bessel function as the reference solution. The Langer/Airy construction is still useful, but in a more targeted role. It gives a local turning-point comparison problem for points where the leading Olver-to-flat map is outside its calibrated gate and where a high-l recurrence would be unnecessarily expensive.

The Airy formula used in that fallback is not just the first-order WKB reference formula. The older Langer formulae used in the comparisons of Refs. [4, 5, 6] are leading uniform Airy approximations: they evaluate an Ai term with the Langer amplitude and action, but do not include the higher Olver correction with a separate Ai′ coefficient. Appendix A derives the second-order one-point version used by the code: the Airy coordinate gives the universal local transition, while the B_0 and A_1 terms account for the first residual correction to the transformed equation. Thus the hierarchy is

$$\text{main path: curved Bessel} \to \text{flat spherical Bessel at } z(\chi),$$

$$\text{high-}l \text{ fallback: curved Bessel} \to \text{second-order Airy at } \zeta(\chi),$$

followed by exact recurrence only outside both calibrated approximation regions. This is also why the shifted-ν limit is tied to the main Olver-to-flat map rather than to the Airy fallback: in the local near-flat regime, the spherical-Bessel coordinate map becomes almost a constant rescaling of the flat argument.

17 Relation to the reference papers

The latest CAMB implementation is closest to the earlier work in its definitions and constraints, but differs in the main approximation used for production:

Reference	Main relevant idea	CAMB relation
Kosowsky, arXiv:astro-ph/9805173 [4]	First-order uniform WKB/Langer approximation for hyperspherical Bessel functions, with analytic action integrals and closed-space symmetry handling.	CAMB uses the same reduced equation and turning-point action ideas. The main path maps to a flat spherical Bessel comparison solution; the Airy comparison reappears only in the second-order fallback, which adds the Ai' correction absent from the leading Langer formula.
Tram, arXiv:1311.0839 [5]	Stable recurrence algorithm, continued-fraction logarithmic derivative, and the closed-space Gegenbauer identity used to cure the backward-recursion obstruction.	`phi_recurs` uses the same recurrence structure and a direct Gegenbauer Miller start as a fallback/start mechanism.
Lesgourgues–Tram, arXiv:1312.2697 [6]	Non-flat CLASS transfer implementation, interpolation strategy, curved-space Limber discussion, and a high-ν flat-rescaling approximation.	CAMB shares the goal of avoiding repeated expensive hyperspherical evaluations and ensuring a smooth flat limit, but uses pointwise Olver-to-flat mapping plus local F, D, F_0 near-flat fillers rather than the CLASS γ, A^l_ν flat-rescaling model.

18 Minimal dictionary between mathematics and code

Mathematical quantity	Code name or code path
$S_K(\chi)$	`State\%rofChi(chi)` or `curved_radius`
u^ν_l	`u_olver` / Numerov variable `y1`
$\phi^\nu_l = u/S_K$	`phi_olver`; in source integration `phi=y1/sh`
$Q = l(l+1)/S_K^2 - \nu^2$	`q_now`, `q_prev`, `q_next` in Numerov stepping
$z(\chi), A(\chi)$	`compute_olver_z_amp` / `compute_olver_z_amp_smallchi`
$\nu_{\rm eff} = \nu F_0$, $\nu_{\rm amp} = \sqrt{F_0}$	`F0`, `nu_eff`, `nu_amp` in near-flat shifted-ν paths
F, D small-χ map	`F0,F2,F4,D0,D2,D4` polynomial coefficients
Flat Bessel lookup	`BessRanges`, `bessel_horner`, `BJL`
Second-order Airy fallback	high-l pointwise fallback before `phi_recurs`
Recursive fallback	`phi_recurs`
Closed Miller start	`phi_closed_gegenbauer_start`

19 Summary

The current non-flat Bessel calculation combines a leading-order Olver-to-flat map, a second-order Airy fallback, a rare open small-ν fallback, exact recurrence fallbacks, and source-integration accelerations. The main map sends the reduced equation $u'' + [\nu^2 - l(l+1)/S_K^2]u = 0$ to the flat comparison solution $zj_l(\nu z)$, with the Jacobian amplitude above, so it is uniform through the turning point and exact for $K = 0$. The Airy branch is a separate high-l fallback, derived in Appendix A, for points where the leading Olver-to-flat map is outside its calibrated gate. In source integration, CAMB uses these point approximations mainly to seed Numerov stepping, while near-flat ranges can be filled from flat Bessel tables. The shifted-ν approximation is the χ-independent part of the local Olver-to-flat expansion, $\nu_{\rm eff} = \nu F_0$, with the simpler $q^2 = \nu^2 - Kl(l+1)/3$ formula retained only as a first-order explanatory limit.

A Second-order Olver Airy approximation

The main text reserves "the Olver approximation" for the default Olver-to-flat map, where the reference problem is the ordinary spherical Bessel equation. This appendix describes the separate Langer/Airy comparison used as a high-l fallback in the gates described above. It is still an Olver uniform approximation, but with an Airy reference equation rather than a flat spherical-Bessel reference equation. The older hyperspherical-Bessel WKB formulae used the leading Langer form, proportional to Ai only. The fallback described here keeps the next Olver terms, producing both an Ai$'$ admixture and a scalar correction to the Ai amplitude. The aim is to show what these second-order terms mean mathematically before describing how they are evaluated cheaply in the one-point implementation.

A.1 Langer form of the radial equation

Start again from the reduced equation

$$u'' = \left[\frac{l(l+1)}{S_K^2(\chi)} - \nu^2\right] u. \tag{106}$$

For an Airy expansion it is convenient to use the standard Langer large parameter[2]

$$\lambda = l + \frac{1}{2}, \qquad \beta = \frac{\nu}{\lambda}. \tag{107}$$

Since $l(l+1) = \lambda^2 - 1/4$, equation (106) becomes

$$u'' = \left[\lambda^2 q(\chi) + r(\chi)\right] u, \qquad q(\chi) = \frac{1}{S_K^2(\chi)} - \beta^2, \qquad r(\chi) = -\frac{1}{4S_K^2(\chi)}. \tag{108}$$

The $q(\chi)$ in this appendix is the local Airy potential, not the dimensional transfer-grid wavenumber q used elsewhere in CAMB. This is the canonical situation for an Olver Airy approximation: there is a large parameter λ, a main potential q, and a smaller residual potential r. The turning point is the zero of q,

$$S_K(\chi_t) = \frac{1}{\beta} = \frac{\lambda}{\nu}. \tag{109}$$

On the origin side of this point $q > 0$, so the regular solution is exponentially small for large l; on the other side $q < 0$, and the solution oscillates. The Airy function is the simplest special function with exactly this behaviour.

A.2 The Airy coordinate

Define the Airy coordinate ζ by matching the action from the turning point:

$$\frac{2}{3}\zeta^{3/2} = \int_\chi^{\chi_t} \sqrt{q(s)}\,\mathrm{d}s, \qquad \chi < \chi_t, \tag{110}$$

and

$$\frac{2}{3}(-\zeta)^{3/2} = \int_{\chi_t}^{\chi} \sqrt{-q(s)}\,\mathrm{d}s, \qquad \chi > \chi_t. \tag{111}$$

Thus $\zeta > 0$ on the evanescent side, $\zeta = 0$ at the turning point, and $\zeta < 0$ on the oscillatory side. Differentiating either branch gives the local identity

$$\left(\frac{\mathrm{d}\zeta}{\mathrm{d}\chi}\right)^2 = \frac{q(\chi)}{\zeta}. \tag{112}$$

[2]This λ is local to the Airy appendix and is unrelated to the Gegenbauer parameter $\lambda = l + 1$ used in the main-text discussion of the closed-space Miller start. The corresponding ratio $\beta = \nu/\lambda$ is also local to this appendix; the main text used $\alpha = \nu/\sqrt{l(l+1)}$ for the leading Olver map.

The ratio q/ζ remains positive on both sides of the turning point, because q and ζ change sign together.

Now make the Liouville transformation

$$u(\chi) = \left(\frac{\zeta}{q}\right)^{1/4} W(\zeta), \tag{113}$$

where the prefactor is understood as its continuous real limit through the turning point. The transformed equation is

$$\frac{\mathrm{d}^2 W}{\mathrm{d}\zeta^2} = \left[\lambda^2 \zeta + \psi(\zeta)\right] W. \tag{114}$$

The leading term is exactly the Airy equation. The function $\psi(\zeta)$ is the residual left over after the action and amplitude have been matched:

$$\psi(\zeta) = \frac{\zeta r}{q} + \frac{\zeta q''}{4q^2} - \frac{5\zeta (q')^2}{16 q^3} + \frac{5}{16\zeta^2}. \tag{115}$$

Here primes denote derivatives with respect to χ, evaluated at the χ corresponding to ζ. For the hyperspherical problem these derivatives are explicit:

$$q' = -\frac{2C_K}{S_K^3}, \qquad q'' = \frac{6}{S_K^4} - \frac{4K}{S_K^2}, \qquad r = -\frac{1}{4S_K^2}. \tag{116}$$

Equation (115) looks singular at $\zeta = 0$, but the singular pieces cancel. The finite turning-point limit is

$$\psi_t \equiv \psi(0) = \frac{2^{1/3}(K - 4\beta^2)(4K - \beta^2)}{280\,\beta^{4/3}(\beta^2 - K)^{4/3}}. \tag{117}$$

This analytic limit is important numerically, because evaluating (115) directly very close to the turning point would subtract large nearly equal terms.

A.3 The second-order Airy expansion

If ψ were zero, equation (114) would have the decaying solution

$$W(\zeta) = \mathrm{Ai}(\lambda^{2/3}\zeta). \tag{118}$$

The second-order Olver expansion corrects this Airy solution by allowing both an Ai coefficient and an Ai$'$ coefficient to vary slowly with ζ. Let

$$X = \lambda^{2/3}\zeta, \qquad Y(X) = \mathrm{Ai}(X), \tag{119}$$

where Y' below denotes differentiation with respect to the Airy argument X. Write

$$W(\zeta) = Y(X)\left[1 + \frac{A_1(\zeta)}{\lambda^2}\right] + \frac{Y'(X)}{\lambda^{4/3}} B_0(\zeta) + \cdots. \tag{120}$$

The powers of λ are fixed by the Airy scaling: $\mathrm{d}/\mathrm{d}\zeta$ acting on $Y(\lambda^{2/3}\zeta)$ brings down a factor $\lambda^{2/3}$, and $Y'' = XY$.

Substituting (120) into (114), then matching the independent Y and Y' pieces order by order, gives

$$2\zeta B_0' + B_0 = \psi, \tag{121}$$

and

$$2A_1' + B_0'' = \psi B_0. \tag{122}$$

Regularity at the turning point fixes the solution of the first equation:

$$B_0(\zeta) = \frac{1}{2\sqrt{\zeta}} \int_0^\zeta \frac{\psi(v)}{\sqrt{v}}\,\mathrm{d}v. \tag{123}$$

For $\zeta < 0$ this is best read in the equivalent real form

$$B_0(\zeta) = \frac{1}{2} \int_0^1 \psi(t\zeta) t^{-1/2}\,\mathrm{d}t. \tag{124}$$

The convention used here is $A_1(0) = 0$, so

$$A_1(\zeta) = \frac{1}{2} \int_0^\zeta \left[\psi(v) B_0(v) - B_0''(v)\right] \mathrm{d}v. \tag{125}$$

Different choices of the constant part of A_1 are possible; they simply move a scalar factor between the coefficient functions and the overall normalization. The code uses the convention (125), with the corresponding normalization correction described below.

Combining the Liouville prefactor and the coefficient functions gives the second-order one-point approximation

$$u_l^\nu(K,\chi) \simeq \sigma \mathcal{N} \left(\frac{\zeta}{q}\right)^{1/4} \left[\mathrm{Ai}(X)\left(1 + \frac{A_1(\zeta)}{\lambda^2}\right) + \frac{B_0(\zeta)}{\lambda^{4/3}}\,\mathrm{Ai}'(X)\right], \qquad X = \lambda^{2/3}\zeta. \tag{126}$$

The sign σ is one except for the closed-space foldings described in the main text. The unreduced hyperspherical Bessel function is then

$$\phi_l^\nu(K,\chi) \simeq \frac{u_l^\nu(K,\chi)}{S_K(\chi)}. \tag{127}$$

A.4 Normalization at the origin

The Airy solution $\mathrm{Ai}(\lambda^{2/3}\zeta)$ selects the solution that decays on the origin side of the turning point, but it does not by itself fix the normalization. The exact regular small-χ behaviour is

$$u_l^\nu(K,\chi) \sim u_0\,\chi^{l+1}, \qquad u_0 = \frac{\prod_{j=1}^l \sqrt{\nu^2 - Kj^2}}{(2l+1)!!}. \tag{128}$$

Near the origin the action has the form

$$\int_\chi^{\chi_t} \sqrt{q(s)}\,\mathrm{d}s = -\log\chi + q_0 + \mathcal{O}(\chi^2), \tag{129}$$

with

$$q_0 = \begin{cases} \log\left(\dfrac{2}{\sqrt{\beta^2-1}}\right) - \dfrac{\beta}{2}\log\left(\dfrac{1+\beta^{-1}}{1-\beta^{-1}}\right), & K = +1, \\ \log\left(\dfrac{2}{\beta}\right) - 1, & K = 0, \\ \log\left(\dfrac{2}{\sqrt{\beta^2+1}}\right) - \beta\arctan(\beta^{-1}), & K = -1. \end{cases} \tag{130}$$

Using the large-positive-X asymptotic form of $\mathrm{Ai}(X)$, the normalization needed in (126) is

$$\mathcal{N} = 2\sqrt{\pi}\,\lambda^{1/6} u_0\, e^{\lambda q_0} R, \tag{131}$$

where R contains the scalar higher-order correction used with the $A_1(0) = 0$ convention:

$$\log R = \frac{1}{\lambda}\left[\frac{1}{12} - \frac{K}{24(\beta^2 - K)}\right] - \frac{1}{225\lambda^2} - \frac{1}{360\lambda^3} + \frac{1}{1260\lambda^5}. \tag{132}$$

For $K = 0$ the curvature term in the square brackets vanishes. This normalization is evaluated in logarithmic form in the implementation so that large l, very small tails, and closed-space products do not overflow or underflow prematurely.

A.5 How the optimized implementation evaluates the second-order terms

The optimized one-point code follows equation (126) directly, but avoids any expensive global construction of B_0 and A_1.

First it folds closed-space χ into the fundamental interval, keeping the exact parity sign σ. It then forms $\lambda = l + 1/2$, $\beta = \nu/\lambda$, the turning point χ_t, the action coordinate ζ, the local value $q(\chi)$, and the Airy argument $X = \lambda^{2/3}\zeta$. The Liouville amplitude is $(\zeta/q)^{1/4}$, with the turning-point limit used when either quantity is too small for the direct ratio.

The second-order coefficient calculation is local to the requested point. The code samples ψ along the same side of the turning point as the requested χ, using the segment

$$\chi_i = \chi_t + \tau_i(\chi - \chi_t), \qquad \tau_i = \{0,\ 0.1464466094,\ 0.5,\ 0.8535533906,\ 1\}\,. \tag{133}$$

At each sample point it computes ζ_i from the analytic action and then evaluates $\psi(\zeta_i)$ from (115). Very near the turning point, $|\zeta| \leq 2\times10^{-3}$, it uses the finite value ψ_t from (117) instead of the subtractive formula. If the requested $|\zeta|$ is below 1.6×10^{-2}, the fit segment is enlarged to this fixed magnitude on the same side of the turn; this keeps the polynomial fit well conditioned while still evaluating the final coefficients at the requested ζ.

The sampled values are converted to a degree-four polynomial in a scaled coordinate,

$$\psi(v) \simeq \sum_{i=0}^{4} c_i \left(\frac{v}{\zeta_s}\right)^i, \tag{134}$$

where normally $\zeta_s = \zeta$, with a small fixed lower magnitude used close to the turning point. The interpolation uses Newton divided differences, which is cheaper than a dense linear solve for this fixed small degree. Because the Olver functionals (124) and (125) are integrals of ψ, the polynomial makes B_0 and A_1 analytic sums. With $\rho = \zeta/\zeta_s$, and for $\zeta \neq 0$,

$$B_0(\zeta) \simeq \sum_{i=0}^{4} \frac{c_i\rho^i}{2i+1}, \tag{135}$$

and

$$A_1(\zeta) \simeq \frac{1}{2}\left[\zeta\sum_{i=0}^{4}\sum_{j=0}^{4}\frac{c_ic_j\rho^{i+j}}{(2j+1)(i+j+1)} - \frac{1}{\zeta}\sum_{i=2}^{4}\frac{i\,c_i\rho^i}{2i+1}\right]. \tag{136}$$

At the turning point the limiting convention is used instead:

$$B_0(0) = \psi_t, \qquad A_1(0) = 0. \tag{137}$$

Thus the apparent $1/\zeta$ in (136) is not evaluated at $\zeta = 0$; it is only the finite-ζ polynomial form of the integral (125). This turning-point coefficient limit is separate from the Liouville-amplitude limit $(\zeta/q)^{1/4}$. For the fitted polynomial approximation to ψ, these sums evaluate the Olver functionals analytically. The remaining error is the local interpolation error in ψ. No inverse $\zeta \mapsto \chi$ solve and no whole-window residual fit is needed.

The final value is then obtained by evaluating $\mathrm{Ai}(X)$ and $\mathrm{Ai}'(X)$, forming the bracket in (126), multiplying by the logarithmic normalization $\mathcal{N}$ and the Liouville amplitude, and dividing by $S_K(\chi)$ if the unreduced ϕ_l^ν is requested. The acceptance gates are deliberately simple. The branch is considered only for $l \geq 10$, positive ν, and nonzero χ. Flat models then pass this base gate. Open models require $\nu \geq 7$, excluding the low-frequency regime where the asymptotic normalization and local residual fit are no longer the best cheap approximation. Closed models require $\beta > 1$ and $\lambda(\beta^2 - 1) \geq 15$, keeping the calculation away from the near-degenerate endpoint where the closed turning point approaches the pole and recurrence is more reliable. There is no symmetric high-$|X|$ acceptance cut. Instead, when $X > 25.77$ the code returns zero, since this is the exponentially small positive Airy tail and is negligible on the peak-normalized accuracy scale. Outside the remaining gates the caller can use the other approximations or the stable recurrence fallback.

Relative to the leading Airy approximation, the new content is therefore not the Airy action map itself, but the local second-order residual correction: $B_0(\zeta)$ supplies the Ai' admixture, and $A_1(\zeta)$ supplies the next scalar correction to the Ai amplitude. The implementation keeps just enough local information about ψ to evaluate these two functions at the single point being requested.

B Open-space small-ν approximation

There is one open-space corner where the usual high-frequency Olver map is not the most efficient description. For $K = -1$, when ν is small compared with the angular barrier, the turning point $\sinh \chi_t = \sqrt{l(l+1)}/\nu$ moves far out in the open tail. The radial equation then spends a long range in a slowly varying exponential barrier. This appendix describes the special small-ν approximation used for that regime.

B.1 Macdonald comparison solution

For open models the reduced equation is

$$u'' + \left[\nu^2 - \frac{l(l+1)}{\sinh^2 \chi}\right] u = 0. \tag{138}$$

Let

$$a_l = \sqrt{l(l+1)}, \qquad b = |\nu| . \tag{139}$$

The symbol a_l is local to this appendix. Unless otherwise stated, the formulae below are written for $b > 0$; the implementation uses the corresponding K_0 and small-b limits when b is zero or extremely small. At large χ, $a_l^2/\sinh^2 \chi \simeq 4a_l^2 e^{-2\chi}$, so the tail of (138) resembles an exponential-barrier equation. The comparison equation is solved by a modified Bessel, or Macdonald, function of imaginary order. Define the scaled real function

$$\mathcal{K}_b(x) = \left[\frac{\sinh(\pi b)}{\pi b}\right]^{1/2} K_{ib}(x), \tag{140}$$

with the limiting $K_0(x)$ interpretation as $b \to 0$. The simplest small-ν approximation is then

$$u_l^\nu(-1,\chi) \simeq \mathcal{K}_b(2\Lambda_{l\nu} e^{-\chi}), \qquad \phi_l^\nu(-1,\chi) \simeq \frac{u_l^\nu(-1,\chi)}{\sinh \chi}. \tag{141}$$

Here $\Lambda_{l\nu}$ is a phase-matching constant. It is fixed by the large-χ phase. As $x \to 0$,

$$\mathcal{K}_b(x) \sim -\frac{1}{b} \sin\left[b \log\left(\frac{x}{2}\right) - \arg\Gamma(1+ib)\right]. \tag{142}$$

The exact open hyperspherical Bessel has the asymptotic phase

$$u_l^\nu(-1,\chi) \sim \frac{1}{b} \sin\left[b\chi - \sum_{j=1}^{l} \arctan\left(\frac{b}{j}\right)\right]. \tag{143}$$

Using

$$\arg\Gamma(l+1+ib) = \arg\Gamma(1+ib) + \sum_{j=1}^{l} \arctan\left(\frac{b}{j}\right), \tag{144}$$

where the argument is the continuous branch obtained by starting at $\arg\Gamma(1) = 0$ and increasing $b \geq 0$, not a principal value folded back into $(-\pi, \pi]$. With this branch choice, the comparison phase matches (143) if

$$\log \Lambda_{l\nu} = \frac{\Im \log \Gamma(l + 1 + ib)}{b}. \tag{145}$$

This expression has a smooth $b \to 0$ limit:

$$\log \Lambda_{l0} = H_l - \gamma, \tag{146}$$

where H_l is the harmonic number and γ is Euler's constant. For small $b/(l+1)$, the branch-continuous expansion used in the code is

$$\log \Lambda_{l\nu} = H_l - \gamma + \sum_{m=1}^{\infty} \frac{(-1)^{m+1} b^{2m}}{2m+1} \sum_{n=l+1}^{\infty} \frac{1}{n^{2m+1}}. \tag{147}$$

This is just the small-b expansion of the gamma-function phase, written as a rapidly convergent tail sum. It is also the branch-continuous route used when $b/(l+1)$ is small, avoiding any ambiguity from jumps of a principal complex logarithm.

B.2 Action-matched argument

Equation (141) matches the asymptotic phase, but its effective exponential barrier is still only an approximation to $a_l^2/\sinh^2\chi$. The optimized version keeps the same comparison function $\mathcal{K}_b$, but replaces the argument $2\Lambda_{l\nu}e^{-\chi}$ by an action-matched argument $x_*(\chi)$. The formulae involving $1/b$ and the comparison actions below are again understood for $b > 0$; the code switches to the limiting K_0 and small-b expressions when b falls below its zero threshold.

The useful variable for the exact open problem is

$$y = \frac{a_l}{\sinh \chi}. \tag{148}$$

Then $\mathrm{d}\chi = -\mathrm{d}y/[y\sqrt{1 + y^2/a_l^2}]$, and the exact WKB action from the turning point $y = b$ is

$$S_{\text{osc}}(y) = \int_y^b \frac{\sqrt{b^2 - t^2}}{t\sqrt{1 + t^2/a_l^2}}\,\mathrm{d}t, \qquad 0 \leq y \leq b, \tag{149}$$

on the oscillatory side, and

$$S_{\text{forb}}(y) = \int_b^y \frac{\sqrt{t^2 - b^2}}{t\sqrt{1 + t^2/a_l^2}}\,\mathrm{d}t, \qquad y \geq b, \tag{150}$$

on the forbidden side. These integrals have stable closed forms. For $0 \leq y \leq b$, with $s = \sqrt{b^2 - y^2}$,

$$S_{\text{osc}}(y) = b \,\operatorname{atanh}\left[\frac{s}{b\sqrt{1 + y^2/a_l^2}}\right] - a_l \operatorname{asin}\left[\frac{s}{\sqrt{a_l^2 + b^2}}\right]. \tag{151}$$

For $y \geq b$, with $s = \sqrt{y^2 - b^2}$,

$$S_{\text{forb}}(y) = a_l \,\operatorname{atanh}\left[\frac{s}{\sqrt{a_l^2 + y^2}}\right] - b \operatorname{asin}\left[\frac{a_l s}{y\sqrt{a_l^2 + b^2}}\right]. \tag{152}$$

The comparison Macdonald equation has the simpler actions

$$C_{\text{osc}}(x) = \int_x^b \frac{\sqrt{b^2 - t^2}}{t}\,\mathrm{d}t = b \log\left(\frac{b + \sqrt{b^2 - x^2}}{x}\right) - \sqrt{b^2 - x^2}, \tag{153}$$

for $0 < x \le b$, and

$$C_{\rm forb}(x) = \int_b^x \frac{\sqrt{t^2 - b^2}}{t}\,\mathrm{d}t = \sqrt{x^2 - b^2} - b \arccos\left(\frac{b}{x}\right), \tag{154}$$

for $x \ge b$. Equations (153) and (154) are the $b > 0$ forms; the same limiting convention described above is used for the very small-b implementation.

The additive constant is chosen so that the action-matched argument has the same far-tail limit as the phase-matched approximation:

$$x_*(\chi) \longrightarrow 2\Lambda_{l\nu} e^{-\chi}, \qquad \chi \to \infty. \tag{155}$$

This gives

$$c_\infty = b\left[\frac{1}{2}\log(a_l^2 + b^2) - \log \Lambda_{l\nu}\right] - b + a_l \arctan\left(\frac{b}{a_l}\right). \tag{156}$$

Define the signed target action

$$T(y) = \begin{cases} S_{\rm osc}(y) + c_\infty, & y < b, \\ -S_{\rm forb}(y) + c_\infty, & y \ge b. \end{cases} \tag{157}$$

The argument x_* is then the point in the comparison problem with the same signed action:

$$C_{\rm osc}(x_*) = T, \qquad T \ge 0, \tag{158}$$

and

$$C_{\rm forb}(x_*) = -T, \qquad T < 0. \tag{159}$$

The final action-matched approximation is

$$u_l^\nu(-1,\chi) \simeq \mathcal{K}_b(x_*(\chi)), \qquad \phi_l^\nu(-1,\chi) \simeq \frac{\mathcal{K}_b(x_*(\chi))}{\sinh\chi}. \tag{160}$$

For the original calibrated range no extra Liouville amplitude is applied. The code does not multiply $\mathcal{K}_b(x_*)$ by a square-root Jacobian built from $\mathrm{d}x_*/\mathrm{d}\chi$, or by a ratio of exact and comparison WKB momenta. It is an action-matched argument replacement, with the normalization fixed by the phase-matched Macdonald tail. For the extended high-l range now used only in the normalized fallback wrapper, the code does multiply by the leading slowly varying factor

$$A_0(\chi) = \sqrt{\frac{\operatorname{asinh}(\operatorname{csch}\chi)}{\operatorname{csch}\chi}}, \tag{161}$$

while deliberately omitting the next $(\nu/\sqrt{l(l+1)})^2$ refinement, which worsened the calibrated mid-l gate. This high-l amplitude factor is not the full x_* Jacobian transformation; it is only a validated extension of the post-Airy open fallback.

B.3 Scaled $K_{i\nu}$ evaluation

The numerical problem is not only to choose x_*, but also to evaluate $\mathcal{K}_b(x)$ without forming a huge $e^{\pi b/2}$ factor times a tiny Macdonald function. The implementation uses the identity

$$\mathcal{K}_b(x) = -\frac{1}{b}\Im\left[e^{i\theta}\sum_{k=0}^{\infty}\frac{(x^2/4)^k}{k!\,(1+ib)_k}\right], \qquad \theta = b\log\left(\frac{x}{2}\right) - \arg\Gamma(1+ib), \tag{162}$$

where $(1+ib)_k$ is the rising Pochhammer symbol. The normalization in (140) cancels the modulus of $\Gamma(1+ib)$, leaving only the phase θ. For very small x the series is replaced by its leading term, which is the small-x phase formula (142).

For $b \gtrsim 20$ the power series becomes cancellation-prone near the Macdonald turning point $x \simeq b$. The code then uses a leading uniform Airy approximation to the Macdonald function. Write $x = bz$. Define a local Macdonald Airy coordinate ζ_K, unrelated to the hyperspherical Airy coordinate in Appendix A, by

$$\frac{2}{3}\zeta_K^{3/2} = \operatorname{atanh}\sqrt{1-z^2} - \sqrt{1-z^2}, \qquad z < 1, \tag{163}$$

and

$$\frac{2}{3}(-\zeta_K)^{3/2} = \sqrt{z^2-1} - \arctan\sqrt{z^2-1}, \qquad z > 1. \tag{164}$$

Thus $\zeta_K > 0$ on the oscillatory side of K_{ib} and $\zeta_K < 0$ on the decaying side. With

$$\Phi_K(z) = \left(\frac{4\,|\zeta_K|}{|1-z^2|}\right)^{1/4}, \qquad \Phi_K(1) = 2^{1/3}, \tag{165}$$

the leading approximation used by the router is

$$\mathcal{K}_b(bz) \simeq \exp\left[\frac{1}{2}\log\left(\frac{\sinh(\pi b)}{\pi b}\right) + \log\pi - \frac{\pi b}{2} - \frac{1}{3}\log b\right]\Phi_K(z)\ \mathrm{Ai}\left(-b^{2/3}\zeta_K\right). \tag{166}$$

For $z < 1$ the Airy argument is negative and oscillatory; for $z > 1$ it is positive and exponentially decaying.

B.4 Implementation details and gates

The small-ν branch is open-space only. Define

$$\nu_0(l) = \max\left[1.2\times 10^{-4}\, l^{3/2},\ \min\left(12\left(\frac{l}{1000}\right)^3,\ 0.032\,l\right)\right]. \tag{167}$$

The intended peak-normalized validity region is

$$l > 20, \qquad \nu \le \begin{cases} \nu_0(l), & l < 3000,\\ \max\left[\nu_0(l),\ 0.04\,l,\ \min\left(8\left(\frac{l}{1000}\right)^2,\ 0.16\,l\right)\right], & l \ge 3000. \end{cases} \tag{168}$$

Outside this calibrated corner the ordinary Olver, Airy, or recurrence paths are preferable. In the current pointwise hierarchy the Airy fallback is tried first, so this small-ν branch is mainly a rare post-Airy safety net, especially at high l.

The implementation first computes $\log\Lambda_{l\nu}$. For $b < 10^{-8}$ it uses $H_l - \gamma$. For $l \ge 8$ and $b/(l+1) < 0.30$, it evaluates the tail series (147) with Euler–Maclaurin estimates for the zeta tails, keeping terms through $m = 4$, and adding the $m = 5, 6$ terms when $b/(l+1) \ge 0.20$; otherwise it evaluates $\arg\Gamma(1+ib) + \sum_{j=1}^{l}\arctan(b/j)$ directly with a real Lanczos phase calculation on the same continuous branch $\arg\Gamma(1) = 0$. The action-matched argument uses $y = a_l/\sinh\chi$, or $y \simeq 2a_l e^{-\chi}$ for very large χ, and uses stable endpoint forms of (151)–(152). The inverse comparison actions (158) and (159) are solved by a few safeguarded Halley steps using small-action and tail asymptotic starting values. If x_* underflows to zero, the code falls back to the non-action small-x phase form, which is the same asymptotic limit.

The scaled Macdonald evaluator is also routed by regime. It works with $\log(x/2)$ whenever possible. For $b < 10^{-8}$ it evaluates the limiting K_0 form. If $\log(x/2) > 120$, it returns zero; if $\log(x/2) < -20$, it uses the leading small-x phase. For $b < 20$, it keeps the older conservative route: the I_{ib} series for $x \le \min[20, \max(3, 0.85b+1)]$, a large-$x$ asymptotic expansion when

$x > 45 + 4b^2$, and otherwise a robust real-axis quadrature of the defining K_{ib} integral. For $b \geq 20$, it uses the checked series while

$$Q_{\text{ser}} \equiv \frac{x^2}{4b} \leq 25 \tag{169}$$

and the terms converge, with at most 384 terms and a squared-term tolerance of 4×10^{-32}; once this test fails, or when $x > b + 5$, it switches to the Airy formula (166). This keeps the fast series in the region where it is accurate, but avoids its near-turning cancellation.

References

[1] F. W. J. Olver, *Asymptotics and Special Functions*, Academic Press, 1974; AKP Classics reprint, 1997.

[2] NIST Digital Library of Mathematical Functions, Chapters 9, 10 and 18.

[3] L. F. Abbott and R. K. Schaefer, "A general, gauge-invariant analysis of the cosmic microwave anisotropy," *Astrophysical Journal* 308, 546 (1986).

[4] A. Kosowsky, "Efficient Computation of Hyperspherical Bessel Functions," arXiv:astro-ph/9805173.

[5] T. Tram, "Computation of hyperspherical Bessel functions," *Communications in Computational Physics* 22, 852–862 (2017), arXiv:1311.0839.

[6] J. Lesgourgues and T. Tram, "Fast and accurate CMB computations in non-flat FLRW universes," *Journal of Cosmology and Astroparticle Physics* 09, 032 (2014), arXiv:1312.2697.